

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0792

Borel 138 kV
November 22, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0792. The Interconnection Customer proposes a 148.4 MW wind facility interconnecting at the Borel 138 kV substation in the Kentucky area. The request was received on May 06, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 298 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$98,337,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Weber to Mullberry 138 KV	Fort Rucker to Abajo 138 KV line	150.4	210	49.0
Mullberry to Salem Walk 138 KV L	Fort Rucker to Abajo 138 KV	146.9	220	49.0
Laguna Tap to Lark Ellen 138 KV L	N. Shannon to Hare Substation 138 KV	141.2	215	59.5
Weber Substation to Lark Ellen 138 KV	Hare to N. Shannon 138 KV	140.3	220	59.5
Studio to Laguna Tap 138 KV L	N. Shannon to Hare Substation 138 KV	140.1	213	59.5
Hare to N. Shannon 138 KV	Weber to Lark Ellen Substation 138 KV	139.2	214	63.3
Kettle Substation to Country Club 2 138 KV L	Spatsum Lornex to Mcclellan Ar 138 KV line	121.8	154	11.8
Harvest Power Orlando to Frontier 138 KV	Spatsum Lornex to Latah Jct 138 KV	113.5	304	22.1
Santa Rosa Beach Substation to Tule 138 KV	Davisville Substation to North Appleton 138 KV	111.0	239	6.4
West Mount Vernon to Santa Rosa Beach Substation 138 KV	N Appleton to Davisville 138 KV Line	109.6	254	6.4
Cortez to S 138 KV line	Weber to Lark Ellen 138 KV line	109.5	197	30.2
Yuba City Sta A to Frontier 138 KV line	Mcclellan Ar to Spatsum Lornex 138 KV	109.3	405	24.4
Reynolds Hill to Howland Rd 138 KV line	Salem Walk to Mullberry 138 KV	108.6	222	62.2
Stony Hill to Sample Rd 138 KV	Ft Rucker to Abajo 138 KV	108.6	189	28.7
Yuba City Sta A Substation to Westin-Elgin 138 KV	Mcclellan Ar to Spatsum Lornex 138 KV	108.5	418	24.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Temple Station 243 to Westin-Elgin 138 KV	Westin-Elgin to McClellan Ar 138 KV	108.4	300	15.3
Nsp to West Mount Vernon 138 KV Line	Davisville to Nrth Appleton 138 KV L	108.4	270	6.4
Wissota to Berry St 138 KV L	Davisville to North Appleton 138 KV line	107.3	648	15.5
McClellan Ar to Westin-Elgin 138 KV line	Yuba City Sta A Substation to Westin-Elgin Substation 138 KV L	107.2	359	19.2
Wissota to Nsp 138 KV L	Davisville to North Appleton 138 KV line	107.2	285	6.4
Granite City Works to Tule 138 KV line	Davisville to North Appleton 138 KV	107.1	240	6.4
Abajo to Hill Country 35 138 KV L	Mullberry to Salem Walk 138 KV Line	106.6	252	37.8
Sand Springs to Russellville E 138 KV	Platte City to Albion Rd 138 KV	105.8	333	7.3
Minco to Barrett 230 KV	Barrett to Norwich 500 KV L	105.4	620	7.8
Russellville to Rancho Carmel 138 KV	Albion Road to Platte City 138 KV	105.1	342	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weber to Mullberry 138 KV	Rebuild line with higher-capacity conductor	\$7,840,000	100% (\$7,840,000)
Salem Walk to Mullberry 138 KV	Rebuild line with higher-capacity conductor	\$4,909,000	100% (\$4,909,000)
Laguna Tap to Lark Ellen 138 KV L	Rebuild line with higher-capacity conductor	\$4,738,000	100% (\$4,738,000)
Lark Ellen to Weber 138 KV	Rebuild line with higher-capacity conductor	\$15,010,000	100% (\$15,010,000)
Studio to Laguna Tap 138 KV	Rebuild line with higher-capacity conductor	\$7,650,000	100% (\$7,650,000)
N. Shannon Substation to Hare 138 KV	Rebuild line with higher-capacity conductor	\$5,122,000	100% (\$5,122,000)
Kettle Substation to Country Club 2 138 KV L	Rebuild line with higher-capacity conductor	\$12,093,000	45% (\$5,481,000)
Frontier to Harvest Power Orlando 138 KV Line	Rebuild line with higher-capacity conductor	\$16,581,000	65% (\$10,757,000)
Tule to Santa Rosa Beach 138 KV	Rebuild line with higher-capacity conductor	\$8,010,000	6% (\$483,000)
West Mount Vernon to Santa Rosa Beach Substation 138 KV	Reconductor line	\$1,684,000	6% (\$102,000)
South to Cortez 138 KV	Reconductor line	\$1,381,000	73% (\$1,011,000)
Frontier to Yuba City Sta A 138 KV L	Reconductor line	\$3,128,000	100% (\$3,128,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Reynolds Hill to Howland Rd 138 KV line	Reconductor line	\$16,331,000	100% (\$16,331,000)
Sample Road to Stony Hill 138 KV line	Reconductor line	\$2,242,000	66% (\$1,479,000)
Westin-Elgin to Yuba City Sta A 138 KV	Reconductor line	\$5,578,000	100% (\$5,578,000)
Temple Station 243 to Westin-Elgin 138 KV	Reconductor line	\$951,000	100% (\$951,000)
Nsp to West Mount Vernon 138 KV Line	Reconductor line	\$4,077,000	6% (\$246,000)
Berry St to Wissota 138 KV	Reconductor line	\$251,000	6% (\$15,000)
Mcclellan Ar to Westin-Elgin 138 KV L	Reconductor line	\$2,640,000	100% (\$2,640,000)
Wissota to Nsp 138 KV	Reconductor line	\$17,571,000	6% (\$1,060,000)
Granite City Works to Tule 138 KV line	Reconductor line	\$6,168,000	6% (\$372,000)
Abajo to Hill Country 345 138 KV	Reconductor line	\$2,728,000	100% (\$2,728,000)
Russellville E to Sand Springs 138 KV	Reconductor line	\$3,926,000	6% (\$222,000)
Minco to Barrett 230 KV	Reconductor line	\$4,159,000	6% (\$248,000)
Rancho Carmel to Russellville East 138 KV	Reconductor line	\$4,176,000	6% (\$236,000)
	Total allocated to project		\$98,337,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.